CS347 — Software Engineering

Faculty of Computer Science, MSA University

Software Requirements Specification

DorMsa

University Housing Platform for MSA Students

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1 Introduction

1.1 Purpose of this Document

This document is the Software Requirements Specification (SRS) for the DorMsa housing platform. It is meant to give the development team, the project supervisor, and anyone else involved a clear and complete picture of what the system needs to do and how it should perform. The document covers both functional and non-functional requirements and is intended to serve as the main reference throughout the project lifecycle.

1.2 Scope of this Document

DorMsa is a web platform built to connect MSA University students who are looking for housing with verified property brokers near the October City campus. The system is designed to cover the whole housing process, from registering an account and searching for listings to communicating with brokers, making deposit payments, and managing listings on the admin side.

1.3 Overview

DorMsa gives three different types of users their own interface and set of features. Students and parents can search for housing, view listings, and book properties. Brokers can list and manage their properties. Admins can verify brokers, moderate content, and monitor the platform. The key features include OTP-based phone verification, search filters, WhatsApp and call integration, secure payment processing, and a real-time analytics dashboard.

1.4 Business Context

MSA University (Modern Sciences and Arts University) is a private university located in October City. Every semester, a large number of foreign students need to find housing near the campus. The problem is that there is no dedicated platform for this. Students currently rely on Facebook groups and recommendations from friends, which often leads to scam listings, wasted time, and in some cases financial problems. DorMsa was created to address this gap. The goal is to build a trusted, verified marketplace that is specifically designed for the MSA community.

1.5 Definitions, Acronyms, and Abbreviations

OTP	One-Time Password. A short-lived code sent to a user's phone for identity verification.
FR	Functional Requirement. A description of a specific behavior the system must support.
NFR	Non-Functional Requirement. A quality or constraint on how the system performs.
PCI-DSS	Payment Card Industry Data Security Standard.
ACID	Atomicity, Consistency, Isolation, Durability.
WCAG	Web Content Accessibility Guidelines.
MSA	Modern Sciences and Arts University, October City, Egypt.
SRS	Software Requirements Specification.
API	Application Programming Interface.
HTTPS	HyperText Transfer Protocol Secure.

2 General Description

2.1 Product Functions

At a high level, DorMsa supports the following functions:

- Role-based user registration and OTP phone verification
- Property search with filters for price, gender, room type, and distance from campus
- Direct broker communication via WhatsApp and Call buttons
- Secure deposit payment processing
- Broker identity verification and listing management
- Admin dashboard for moderation, analytics, and user management
- Automated notifications for bookings, new listings, and listing expiry reminders

2.2 Similar System Information

DorMsa is a standalone platform. It integrates with several external systems: the WhatsApp API for direct broker communication, Google Maps for proximity calculations, an SMS gateway (Twilio) for OTP delivery, and a PCI-DSS compliant payment gateway for deposit processing. While platforms like Airbnb and Beytna exist globally, DorMsa is different because it focuses entirely on the MSA student community and includes MSA-specific verification and student-focused filtering.

2.3 User Characteristics

- **Students and Parents:** Primary users, mostly MSA students or families looking for off-campus housing. Most will use the platform on a mobile device and are assumed to have basic smartphone experience but no technical background.
- **Brokers:** Property owners or rental agents who list dorms near MSA campus. Expected to be comfortable with web forms and uploading photos.
- **Admins:** Staff from MSA's IT or housing department responsible for verifying brokers, moderating content, and monitoring the platform.

2.4 User Problem Statement

MSA students currently face several problems when trying to find off-campus housing. There is no centralized platform designed specifically for them. They usually turn to unverified Facebook groups or word-of-mouth, which regularly leads to scam listings. Brokers have no professional way to reach students. There is also no easy way to filter by gender-appropriate housing, and without a formal deposit system, payment disputes are common.

2.5 User Objectives

Students and parents want to find verified, affordable, gender-appropriate housing near MSA as quickly and easily as possible. Brokers want a professional channel to list their properties. Admins want to keep the platform trustworthy by verifying brokers and removing bad content in a timely manner.

2.6 General Constraints

The system must be deployed as a web application and optimized for mobile browsers. All payment processing must comply with PCI-DSS. The system must support at least 1,000 concurrent users from the moment it launches.

3 Functional Requirements

All 49 functional requirements are formatted using the 17-row Stephen Withall structural grid. The Requirement Description cell mirrors the reference pattern, supporting both plain text and bulleted sub-items where applicable.

Requirement Identifier	FR01
Requirement Name	User Registration
Requirement Description	The system shall allow users to register using their name, phone number, and selected role. • Users may register as: Student, Parent, or Broker. • A temporary account is created pending OTP phone verification. • Duplicate phone numbers must be rejected.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	User Management
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR01 (Data Encryption), NFR08 (Registration Speed), NFR26 (Password Complexity), NFR28 (Input Sanitization)
Acceptance Criteria	User can submit registration data and a temporary inactive account state is created pending verification. Duplicate phone numbers are rejected with a clear error message.
Related Requirements	FR02, FR05
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR02
Requirement Name	OTP Verification
Requirement Description	The system shall verify phone numbers via a One-Time Password (OTP) to ensure trusted communication. • A 6-digit OTP is sent to the user's registered phone number. • OTP must be entered within a defined expiry window. • On valid entry the account status changes from inactive to active.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR43 (Low Latency OTP)
Acceptance Criteria	Account status changes from inactive to active upon valid 6-digit OTP entry within the expiry window.
Related Requirements	FR01
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR03
Requirement Name	Secure Login / Logout
Requirement Description	The system shall provide secure login and logout functionality for authenticated access. • Login requires a verified phone number and password. • On logout all active session tokens for that device are invalidated. • Sessions expire automatically after 30 minutes of inactivity (see NFR04).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	User Management
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR04 (Session Timeout), NFR26 (Password Complexity), NFR32 (HTTPS Enforcement)
Acceptance Criteria	Active session tokens are dropped on logout. Login is granted only with correct credentials. Inactive sessions expire after 30 minutes.
Related Requirements	FR02, FR05
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR04
Requirement Name	Profile Editing
Requirement Description	The system shall allow users to update their personal contact details and profile information. • Editable fields include: name, phone number, and profile preferences. • Changes are persisted to the database immediately upon saving. • Input is validated and sanitized before storage.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	User Management
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR28 (Input Sanitization), NFR42 (Font Legibility)
Acceptance Criteria	Modified profile attributes reflect immediately in the database upon save.
Related Requirements	FR03
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR05
Requirement Name	Role-Based Access
Requirement Description	The system shall restrict access to specific features based on the user's assigned role. • Roles: Student/Parent, Broker, Admin. • Each role is directed to a distinct dashboard upon login. • Cross-role access attempts are blocked and logged.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR06 (Access Control)
Acceptance Criteria	Broker interface blocks access to student shortlist panels and vice-versa. Admin dashboard is inaccessible to non-admin users.
Related Requirements	FR01, FR03
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR06
Requirement Name	Password Recovery
Requirement Description	The system shall allow users to reset their credentials via their registered phone number. • User receives an OTP to their registered number to initiate reset. • Password reset is inaccessible without completing OTP validation. • A confirmation message is sent after successful reset.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	User Management
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR43 (Low Latency OTP)
Acceptance Criteria	Password reset is inaccessible until cellular ownership validation is completed. Credentials are updated and a confirmation message is delivered.
Related Requirements	FR02
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR07
Requirement Name	Price Filtering
Requirement Description	The system shall allow users to filter listings based on a minimum and maximum price range. • Users can enter both a minimum and a maximum price bound. • Listings outside the specified range are excluded from results immediately. • Filter state persists during the same session.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR10 (Page Load Time), NFR47 (Database Indexing)
Acceptance Criteria	Entering price bounds excludes out-of-range records immediately from search results.
Related Requirements	FR21, FR05
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR08
Requirement Name	Room Type Filter
Requirement Description	The system shall allow users to filter by room type (Single, Double, Suite). • Available room types: Single, Double, Suite. • Only listings matching the selected room type are returned. • Multiple room types may be selected simultaneously.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR47 (Database Indexing)
Acceptance Criteria	Database retrieval displays exclusively room schemas matching the structural selection.
Related Requirements	FR21, FR07
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR09
Requirement Name	Gender-Specific Search
Requirement Description	The system shall filter dorms by gender designation (Male, Female, Mixed). • Designation options: Male, Female, Mixed. • Only listings with the matching gender designation are returned. • This filter is prominently exposed on the search screen.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR37 (Search Precision)
Acceptance Criteria	Female students view only female or mixed listings when that filter is applied.
Related Requirements	FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR10
Requirement Name	Proximity Search
Requirement Description	The system shall filter listings based on distance from MSA University campus. • Distance is calculated from the fixed MSA October City campus coordinates. • Results are sorted in ascending order by distance. • Requires Google Maps API integration (see NFR25).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR25 (API Latency), NFR47 (Database Indexing)
Acceptance Criteria	Listings are displayed sorted in ascending order by distance from campus.
Related Requirements	FR16, FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR11
Requirement Name	Property Details View
Requirement Description	The system shall display high-resolution images and full text descriptions for every listing. • Each listing page shows: photo gallery, description, price, room type, gender, location map. • Images are compressed for mobile loading (see NFR09). • Full details load within 2 seconds (see NFR10).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Listings
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR09 (Image Optimization), NFR10 (Page Load Time)
Acceptance Criteria	Full property details, images, and location are rendered correctly within 2 seconds.
Related Requirements	FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR13
Requirement Name	Shortlist / Favourites
Requirement Description	The system shall allow users to save listings to a Shortlist for later comparison. • A heart/bookmark icon on each listing card allows saving. • Saved listings persist across sessions for the logged-in user. • Users can remove listings from their shortlist at any time.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR11 (Intuitive Navigation)
Acceptance Criteria	Listing indices are preserved inside the student dashboard across sessions.
Related Requirements	FR11, FR03
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR14
Requirement Name	WhatsApp Integration
Requirement Description	The system shall provide a direct button to initiate a WhatsApp chat with the broker. • A “Chat on WhatsApp” button is displayed on every verified broker listing. • Clicking the button launches the WhatsApp app with the broker’s verified number pre-filled. • The click event is recorded for inquiry analytics (see FR25).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Communication
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR21 (Cross-Browser Support), NFR07 (Mobile-First Design)
Acceptance Criteria	Clicking the trigger launches a deep-link communication session passing the broker’s verified contact data.
Related Requirements	FR28, FR11
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR15
Requirement Name	Direct Call Button
Requirement Description	The system shall provide a Call button to contact brokers directly via phone. • A “Call Broker” button is displayed on every verified broker listing. • Clicking the button launches the native phone dialer with the broker’s number. • The click event is recorded for inquiry analytics (see FR25).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Communication
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR07 (Mobile-First Design)
Acceptance Criteria	The native phone dialer launches with the broker’s verified number upon button tap.
Related Requirements	FR28
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR16
Requirement Name	Location Map View
Requirement Description	The system shall display the exact location of dorms on an interactive map. • A Google Maps embed or equivalent renders on each listing's detail page. • A map pin is placed at the property's GPS coordinates. • Requires a valid address entered by the broker at listing creation.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR25 (API Latency)
Acceptance Criteria	Map renders with an accurate pin at the property's coordinates without breaking the UI layout.
Related Requirements	FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR17
Requirement Name	Deposit Payment
Requirement Description	The system shall process secure deposit payments to confirm a booking. • Students pay a deposit to secure a listing. • Transactions are routed through a PCI-DSS compliant payment gateway (see NFR05). • A payment receipt is generated and both student and broker are notified.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Finance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR05 (PCI-DSS Compliance), NFR24 (Database Consistency)
Acceptance Criteria	Transactional records are locked and updated atomically. A receipt is generated and notification is dispatched to both parties.
Related Requirements	FR34
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR18
Requirement Name	Rating System
Requirement Description	The system shall allow users to rate brokers and properties after a confirmed visit. • Rating scale: 1 to 5 stars. • Only students who have completed a confirmed booking may rate. • Ratings are published on the broker and property profiles.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Feedback
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR28 (Input Sanitization)
Acceptance Criteria	User scores update system averages accurately within the target database row.
Related Requirements	FR17, FR34
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR19
Requirement Name	Review Submission
Requirement Description	The system shall allow users to write text reviews about their housing experience. • Text reviews are available to students who have completed a stay. • Reviews are stored, sanitized, and displayed publicly on the listing. • Admins may remove reviews that violate platform policies.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Feedback
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR28 (Input Sanitization)
Acceptance Criteria	Text reviews are sanitized for vulnerabilities and displayed under the target listing.
Related Requirements	FR18
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR20
Requirement Name	Saved Searches
Requirement Description	The system shall allow users to save their search criteria for future use. • Users can name and save any combination of active filter parameters. • Saved searches are retrievable from the user's profile in future sessions. • Saved searches trigger new listing alerts (see FR35).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR47 (Database Indexing)
Acceptance Criteria	Complex filter criteria are saved to student profile records and can be recalled in future sessions.
Related Requirements	FR35
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR21
Requirement Name	Create Listing
Requirement Description	The system shall allow verified brokers to upload property details, photos, and locations. • Required fields: title, price, room type, gender designation, address, at least one photo. • Broker must be verified by an Admin before any listing can be created (see FR28). • Uploaded images are scanned for malware before storage (see NFR45).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Inventory
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR19 (Scalable Storage), NFR45 (Secure File Upload)
Acceptance Criteria	Property data structures persist correctly in the database and the listing becomes publicly visible.
Related Requirements	FR28, FR27
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR22
Requirement Name	Edit Listing
Requirement Description	The system shall allow brokers to update prices, descriptions, and photos of active listings. • Brokers may update any field on a listing they own. • Changes are published immediately to all users (see NFR14). • An audit trail entry is created for each edit.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Inventory
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR14 (Data Refresh Rate)
Acceptance Criteria	Mutated field structures are published immediately across all public-facing interfaces.
Related Requirements	FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR23
Requirement Name	Availability Toggle
Requirement Description	The system shall allow brokers to mark a property as Available or Rented. • Toggle states: Available, Rented. • “Rented” listings are hidden from active search results immediately. • The toggle is accessible from the broker dashboard with a single action.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Inventory
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR14 (Data Refresh Rate)
Acceptance Criteria	Toggling the availability state hides or shows the listing in search results instantly.
Related Requirements	FR21, FR22
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR24
Requirement Name	Broker Dashboard
Requirement Description	The system shall provide a dashboard showing view counts and student inquiry metrics. • Metrics displayed: total listing views, WhatsApp clicks, Call clicks, bookings. • Data refreshes in real-time. • Accessible only to the authenticated broker who owns the listings.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Analytics
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR44 (Memory Management)
Acceptance Criteria	Calculation charts generate accurate performance visualizations on access.
Related Requirements	FR21, FR25
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR25
Requirement Name	Inquiry Tracking
Requirement Description	The system shall track the number of times a broker's phone number was clicked. • Every WhatsApp and Call button click is counted and timestamped. • Counts are stored in the broker's analytics record. • Data is surfaced in the broker dashboard (see FR24).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Analytics
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR24 (Database Consistency)
Acceptance Criteria	The audit ledger increments the action count whenever a contact button is triggered.
Related Requirements	FR14, FR15
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR26
Requirement Name	Bulk Management
Requirement Description	The system shall provide tools for brokers to manage multiple listings simultaneously. • Brokers may select multiple listings and apply a single action (e.g. price update, hide). • Requires at least two active listings. • Bulk actions are confirmed before execution to prevent accidental changes.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Inventory
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR10 (Page Load Time)
Acceptance Criteria	Batch modification actions propagate to all selected listings simultaneously.
Related Requirements	FR22, FR23
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR27
Requirement Name	Identity Submission
Requirement Description	The system shall allow brokers to upload identification documents for account verification. • Acceptable document types: national ID, passport. • Documents are stored securely in a restricted repository pending Admin review. • Uploaded files are scanned for malware (see NFR45).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR45 (Secure File Upload)
Acceptance Criteria	Document records are deposited cleanly in a secure repository and become available for Admin review.
Related Requirements	FR28
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR28
Requirement Name	Broker Verification Panel
Requirement Description	The system shall provide a panel for admins to approve or reject broker accounts. • Admins view a queue of brokers awaiting verification. • Admins may approve or reject with an optional reason. • The broker is notified of the outcome by SMS or push notification.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Admin
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR03 (Manual Verification), NFR06 (Access Control)
Acceptance Criteria	Approval toggles grant posting permissions; rejection blocks listing creation. The broker receives an automated notification of the decision.
Related Requirements	FR27, FR05
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR29
Requirement Name	Content Moderation
Requirement Description	The system shall allow admins to delete listings with fake or inappropriate content. • Admins can remove any listing from the platform. • Removed listings are logged with a reason and timestamp. • The broker receives a warning notification upon removal.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Admin
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR37 (Search Precision)
Acceptance Criteria	Destructive administrative actions purge flagged listings from the system index. The broker receives a warning.
Related Requirements	FR37, FR30
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR30
Requirement Name	User Suspension
Requirement Description	The system shall allow admins to ban users reported for misconduct. • Suspended users cannot log in or access the platform. • All active sessions for the suspended user are immediately invalidated. • Suspension is logged in the audit trail (see FR37).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Admin
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR06 (Access Control)
Acceptance Criteria	Active sessions drop instantly and future login attempts fail once the ban flag is set.
Related Requirements	FR37
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR31
Requirement Name	System Usage Analytics
Requirement Description	The system shall generate reports on active listings and registered students. • Reports include: total registered users per role, active listings, bookings per period. • Available only in the Admin analytics portal. • Data is displayed in visual chart form.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Won't
Business Area	Analytics
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR33 (Performance Monitoring)
Acceptance Criteria	Platform metric aggregates compute correctly over broad timeline query selections.
Related Requirements	FR32
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR32
Requirement Name	Price Range Reports
Requirement Description	The system shall generate reports on the most searched price ranges. • Reports rank price ranges by search frequency. • Requires sufficient historical search data to be meaningful. • Accessible only to Admins.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Won't
Business Area	Analytics
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR23 (Log Retention)
Acceptance Criteria	The analytical system lists price range data points sorted by hit frequency.
Related Requirements	FR07, FR31
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR33
Requirement Name	Fraud Detection
Requirement Description	The system shall automatically flag suspicious patterns or duplicate listings. • Background jobs compare listing content to detect duplicates. • Suspicious listings are flagged and surfaced to the Admin review queue. • The detection process runs without user intervention.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR02 (Anti-Scraping)
Acceptance Criteria	Comparison background jobs isolate matching listing strings and populate the Admin alert log.
Related Requirements	FR29, FR37
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR34
Requirement Name	Booking Confirmation
Requirement Description	The system shall send an automated notification when a booking is successful. • Triggered immediately upon successful deposit payment. • Both the student and the broker receive a confirmation notification. • Notification channels: SMS and/or push notification (see FR46).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Must
Business Area	Notifications
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR43 (Low Latency OTP)
Acceptance Criteria	Receipt execution triggers automated communication flows to both the student and broker.
Related Requirements	FR17, FR46
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR35
Requirement Name	New Listing Alert
Requirement Description	The system shall notify users when a new listing matches their saved search criteria. • Requires user to have at least one saved search (see FR20). • Alert is dispatched when a new listing matches saved criteria. • Delivered via SMS or push notification (see FR46).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Notifications
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR46 (API Rate Limiting)
Acceptance Criteria	Listing insertion triggers notification distribution to all users with a matching saved search.
Related Requirements	FR20, FR46
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR36
Requirement Name	Expiry Reminders
Requirement Description	The system shall notify brokers to update listings older than 30 days. • A reminder is sent if a listing has not been modified for 30 days. • Delivered via SMS or push notification. • If no action is taken within another 30 days the listing is auto-hidden (see FR47).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Notifications
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR15 (Backup Frequency)
Acceptance Criteria	System lifecycle loops trace elapsed temporal parameters and dispatch expiry signals to the broker.
Related Requirements	FR47
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR37
Requirement Name	Audit Logs
Requirement Description	The system shall record all admin actions for security and tracking purposes. • Every Admin action (approve, reject, suspend, delete) is logged. • Each log entry includes: action type, target entity, timestamp, Admin ID. • Logs are retained for a minimum of 90 days (see NFR23).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR23 (Log Retention)
Acceptance Criteria	Modification actions persist structured metadata within immutable operational log files.
Related Requirements	NFR23
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR38
Requirement Name	Language Localization
Requirement Description	The system shall support multiple languages for international students. • Initial supported languages: Arabic, English. • Language preference is set from the user's profile settings. • All UI text is rendered in the selected language upon preference change.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR12 (WCAG Compliance), NFR27 (Localization Support)
Acceptance Criteria	Structural text translation maps apply smoothly across all user display panels.
Related Requirements	NFR27
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR39
Requirement Name	Currency Converter
Requirement Description	The system shall provide real-time currency conversion for housing prices. • Users can select a preferred display currency from a dropdown. • Conversion rates are fetched from a live exchange rate API. • Prices update immediately upon currency selection.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR25 (API Latency)
Acceptance Criteria	Mathematical exchange processing maps localized prices on selection.
Related Requirements	FR07
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR40
Requirement Name	Safety Indicators
Requirement Description	The system shall display safety indicators for dorm locations based on local data. • Safety rating or badge is shown on each listing where location data is available. • Data is sourced from verified local safety datasets. • Indicator updates when underlying data changes.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR37 (Search Precision)
Acceptance Criteria	Background metadata attaches visual safety elements to matching geo-location records.
Related Requirements	FR16
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR41
Requirement Name	Amenities Filter
Requirement Description	The system shall filter listings by nearby amenities such as supermarkets. • Amenity categories: supermarket, pharmacy, public transport, gym. • Users select one or more amenity categories in the search filter panel. • Only listings within a defined radius of the selected amenity are shown.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Search
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR10 (Page Load Time)
Acceptance Criteria	Location queries return listings bounded strictly within the contextual amenity buffer radius.
Related Requirements	FR10
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR42
Requirement Name	Roommate Matching
Requirement Description	The system shall suggest potential roommates based on student profile preferences. • Students complete a preferences section in their profile (see FR04). • The system suggests compatible students based on preferences. • Students may opt out of roommate matching visibility from their settings.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Social
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR47 (Database Indexing)
Acceptance Criteria	Correlation routines evaluate compatibility vectors and display matching student matrices.
Related Requirements	FR04
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR43
Requirement Name	Contract Template
Requirement Description	The system shall provide digital contract templates for brokers and students. • Generated automatically following a confirmed booking (see FR34). • Template is pre-filled with booking details: student name, property, dates, deposit amount. • Both parties receive a copy of the generated document.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Legal
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR41 (Legal Compliance)
Acceptance Criteria	Document generators compile user metadata into a legally structured contract template.
Related Requirements	FR17, FR34
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR44
Requirement Name	Payment History
Requirement Description	Users shall be able to view their previous deposit payment transactions. • Each transaction record shows: date, amount, property name, and status. • Records are listed in reverse chronological order. • Accessible from the student's profile dashboard.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Finance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR24 (Database Consistency)
Acceptance Criteria	Lookup routines read transaction fields accurately and list them chronologically.
Related Requirements	FR17
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR45
Requirement Name	TroubleShoot
Requirement Description	The system shall allow users to submit support tickets for technical issues. • Users complete a support form with: issue type, description, and optional screenshot. • A unique ticket ID is generated and communicated to the user. • Tickets are assigned to the support team queue for resolution.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Support
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR22 (Error Message Clarity)
Acceptance Criteria	Processing engines spawn unique issue IDs and map payloads into the administrative support track.
Related Requirements	FR50
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR46
Requirement Name	Push Notifications
Requirement Description	The system shall send push notifications for mobile browser users. • Users must grant notification permission in their browser settings. • Notification types: booking confirmation, new listing alert, expiry reminder. • Delivered via the browser's Web Push API.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Notifications
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR07 (Mobile-First Design)
Acceptance Criteria	The service worker relays alerts instantly to device notification channels for permissioned users.
Related Requirements	FR34, FR35
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR47
Requirement Name	Listing Expiry Auto-Hide
Requirement Description	The system shall automatically hide listings not updated for 60 days. • Automated sweep checks listings daily for last-modification date. • Listings exceeding 60 days without update are hidden from search results. • The broker receives a notification when their listing is auto-hidden.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Inventory
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR14 (Data Refresh Rate)
Acceptance Criteria	Structural visibility flags shift to hidden during automated sweeps after the 60-day threshold.
Related Requirements	FR36, FR23
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR48
Requirement Name	Profile Verification Badge
Requirement Description	The system shall display a verification badge on verified broker profiles. • Badge is displayed on the broker's profile and on all their active listings. • Badge is assigned only after Admin approval (see FR28). • Badge is removed if the broker's account is suspended.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR35 (Consistent Branding)
Acceptance Criteria	Condition checks render the verification badge graphic next to all approved broker display blocks.
Related Requirements	FR28
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR49
Requirement Name	Student ID Verification
Requirement Description	The system shall verify student status via university email or student ID upload. • Acceptable verification: MSA university email confirmation or student ID upload. • Verification is optional but unlocks additional trust features. • Verified student accounts display a student-verified indicator.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Should
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR49 (User Consent)
Acceptance Criteria	Internal state mutations track university validation steps and flag the account as student-verified.
Related Requirements	FR01, FR28
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	FR50
Requirement Name	FAQ Section
Requirement Description	The system shall provide a frequently asked questions section for new users. • FAQ content is accessible without requiring login. • Users can search FAQ articles by keyword. • Admins may add or update FAQ entries from the admin panel.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Functional Requirement (FR)
Priority	Could
Business Area	Support
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	NFR11 (Intuitive Navigation)
Acceptance Criteria	Static FAQ fragments display correctly across unauthenticated portal contexts and are keyword-searchable.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

4 Interface Requirements

4.1 User Interfaces

The platform uses a mobile-first graphical interface. The main screens are:

- **Registration and Login Screen:** includes phone input, role selection, and OTP entry fields.
- **Home / Search Screen:** filter bar for price, gender, room type, and distance, with listing cards.
- **Property Detail Screen:** photo gallery, description, price, location map, Call and WhatsApp buttons.
- **Shortlist Screen:** saved listings grid with options to remove or compare entries.
- **Broker Dashboard:** table for managing listings, toggling availability, and viewing analytics.
- **Admin Panel:** pending broker approvals, flagged listings, and system analytics.
- **Notifications Screen:** timeline of booking confirmations, listing alerts, and expiry reminders.

All screens follow MSA's official colors (gold #8B6914 and navy #1F3864), maintain a minimum body font size of 14px (NFR42), and are designed so any listing is reachable within 3 clicks (NFR11).

4.2 API (Application Programming Interface)

The system exposes a RESTful API for front-end communication:

- POST /api/auth/register — creates a new user account
- POST /api/auth/verify-otp — validates OTP and activates the account
- GET /api/listings?price min=&price max=&gender=&distance= — returns filtered listings
- POST /api/listings — creates a new listing (broker-authenticated)
- POST /api/payments/deposit — initiates a secure deposit payment
- GET /api/admin/pending-brokers — returns the list of unverified brokers
- PATCH /api/admin/brokers/:id/approve — approves a broker account

4.3 Diagnostics

The system logs all errors with stack traces to a centralized logging service. Admins can access an error dashboard to monitor system health. Logs are retained for 90 days (NFR23).

4.4 Hardware Interfaces

DorMsa is entirely web-based. It runs on standard cloud servers with a minimum of 4 vCPU and 8 GB RAM and can be accessed from any modern smartphone or computer with an internet connection.

4.5 Communications Interfaces

HTTPS with TLS 1.3 is used for all client-server communication. OTP delivery uses Twilio SMS. WhatsApp API provides deep links for broker communication. Google Maps API handles geolocation. A PCI-DSS compliant payment gateway handles all deposit transactions.

4.6 Software Interfaces

The backend relies on a MongoDB NoSQL database for storing user, listing, booking, and review data. Booking confirmation emails are sent through a service such as SendGrid.

5 Performance Requirements

The system must meet the following performance targets:

- Search results must load within 2 seconds under normal network conditions (NFR10).
- The system must support at least 1,000 concurrent users without performance degradation (NFR17).
- OTP messages must be delivered within 15 seconds of being requested (NFR43).
- External API calls such as Google Maps must respond within 500 milliseconds (NFR25).
- The mobile application must not exceed 200 MB RAM usage on client devices (NFR44).
- Property images must be optimized to load properly over 3G mobile connections (NFR09).
- The registration process must be completable within 30 seconds (NFR08).
- Frequently searched database fields must be indexed for sub-second query execution (NFR47).

6 Design Constraints

6.1 Standards Compliance

The system must comply with PCI-DSS for payment security, use HTTPS/TLS 1.3, collect user data only with explicit consent, aim for WCAG 2.1 Level AA, follow Egyptian student housing regulations, and ensure backend code follows PEP8 or an equivalent standard.

6.2 Hardware Limitations

The system must work on devices with a screen width of at least 320px. Application RAM usage on mobile must not exceed 200 MB (NFR44). No native device hardware is required.

6.3 Other Constraints

The system must use a modular architecture (NFR18). All source code must be in a Git repository (NFR30). Cloud storage for images must auto-scale during peak upload periods (NFR19).

7 Non-Functional Requirements

All 40 non-functional requirements are formatted using the same 17-row Stephen Withall structural grid used for functional requirements.

Requirement Identifier	NFR01
Requirement Name	Data Encryption
Requirement Description	All user names and phone numbers must be encrypted at rest using AES-256. • Encryption applies to all personally identifiable fields stored in the database. • AES-256 is the minimum accepted encryption standard. • Unencrypted plaintext must never appear in database dumps or logs.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR01, FR03
Acceptance Criteria	Unencrypted plaintext fields are fully absent from database dump variables.
Related Requirements	FR01
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR02
Requirement Name	Anti-Scraping
Requirement Description	The system must implement protections to prevent automated data scraping. • Bot detection mechanisms (e.g. rate limiting, CAPTCHA) are applied to public endpoints. • Automated script access beyond defined thresholds is blocked with a 429 response.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR21, FR11
Acceptance Criteria	Automated script triggers are blocked safely by protection mechanisms.
Related Requirements	FR33
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR03
Requirement Name	Manual Verification
Requirement Description	Broker identities must be manually verified by an Admin before they can post listings. • No listing creation is possible for any broker whose account is not Admin-approved. • The verification process is carried out via the Broker Verification Panel (FR28).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR21, FR28
Acceptance Criteria	Listing submission fails completely for unapproved broker accounts.
Related Requirements	FR28
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR04
Requirement Name	Session Timeout
Requirement Description	User sessions shall automatically expire after 30 minutes of inactivity. • Inactivity is defined as no user interaction within a 30-minute window. • On expiry the session token is invalidated and the user is redirected to login.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR03
Acceptance Criteria	Authentication validation fails once the idle state exceeds 30 minutes.
Related Requirements	FR03
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR05
Requirement Name	PCI-DSS Compliance
Requirement Description	All deposit transactions must comply with PCI-DSS security standards. • Credit card data must never touch local infrastructure; it is handled entirely by the payment gateway. • The selected payment gateway must hold a valid PCI-DSS certification.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR17
Acceptance Criteria	Credit card data never touches local infrastructure during any checkout execution.
Related Requirements	FR17
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR06
Requirement Name	Access Control
Requirement Description	Only authorized Admin staff shall have access to the management dashboard. • Non-admin session tokens are rejected when accessing admin-designated routes. • Admin accounts are provisioned and managed separately from regular user registration.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR05, FR28
Acceptance Criteria	Non-admin session validation drops any requests directed at secure admin links.
Related Requirements	FR05
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR07
Requirement Name	Mobile-First Design
Requirement Description	The interface must be optimized for mobile devices and small screens. • All layouts must render correctly at 320px minimum viewport width. • No horizontal scrolling should appear at standard mobile breakpoints.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR11, FR15
Acceptance Criteria	Viewport variables adapt without horizontal scrolling down to 320px.
Related Requirements	FR15
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR08
Requirement Name	Registration Speed
Requirement Description	The registration process must be completable in under 30 seconds. • Measured from first page load to OTP-verified account activation. • Applies under normal network and server conditions.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR01, FR02
Acceptance Criteria	Total user registration journey completes within 30 seconds during normal trial workflows.
Related Requirements	FR01
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR09
Requirement Name	Image Optimization
Requirement Description	Property images must be compressed and optimized for loading on slow mobile connections. • Images are automatically resized and compressed on upload. • Optimized images are served via a CDN where possible.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR11, FR21
Acceptance Criteria	Processing pipeline outputs resized low-overhead image files automatically.
Related Requirements	FR11
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR10
Requirement Name	Page Load Time
Requirement Description	All search results must load within 2 seconds under normal conditions. • Measured from user submission of search query to full result display. • Applies under standard broadband network conditions and normal server load.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR07, FR10
Acceptance Criteria	Query return parameters settle below 2.0 seconds across benchmark logs.
Related Requirements	FR07
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR11
Requirement Name	Intuitive Navigation
Requirement Description	Users should be able to find any listing within 3 clicks from the home page. • Interaction depth from home page to a specific listing must not exceed 3 steps. • Measured using a standard filter-and-select task flow.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR07, FR11
Acceptance Criteria	Interface interaction depth stays under 3 transition operations from home to any listing.
Related Requirements	FR13
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR12
Requirement Name	WCAG Compliance
Requirement Description	The system should comply with WCAG 2.1 accessibility standards. • Target conformance level: WCAG 2.1 Level AA. • Applies to all public-facing interface views.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All front-end modules
Acceptance Criteria	Context color contrast metrics maintain compliance parameters.
Related Requirements	FR38
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR13
Requirement Name	System Uptime
Requirement Description	The system must maintain 99.9% uptime during peak months (September – November). • Measured over each calendar month during the peak booking period. • Planned maintenance windows must be scheduled outside peak periods.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Availability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All systems
Acceptance Criteria	Deployment infrastructure remains functional across seasonal high-traffic cycles.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR14
Requirement Name	Data Refresh Rate
Requirement Description	Search results must refresh immediately whenever a broker updates a listing. • Cache invalidation is triggered on any listing save or availability toggle. • Changes are visible to searching users within 1 second of the broker saving.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR22, FR23
Acceptance Criteria	Modifications clear internal server-side view caches instantly on event changes.
Related Requirements	FR22, FR23, FR47
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR15
Requirement Name	Backup Frequency
Requirement Description	System data must be backed up automatically every 24 hours. • Backups are encrypted and stored in a separate cloud region. • Backup success or failure generates an automated alert to the Admin.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Database system
Acceptance Criteria	Scheduled handlers write encrypted database snapshot files daily to cloud storage.
Related Requirements	FR36
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR16
Requirement Name	Disaster Recovery
Requirement Description	The system must be restorable within 4 hours of a critical failure. • Recovery Time Objective (RTO): 4 hours. • Recovery procedure must be documented and regularly tested.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All systems
Acceptance Criteria	Cold restore runs achieve operational stability within 4 hours.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR17
Requirement Name	Concurrent Users
Requirement Description	The system must support at least 1,000 concurrent users without degradation. • Load tests must demonstrate stability at 1,000 simultaneous sessions. • Performance metrics (response time, error rate) must remain within defined thresholds under load.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All frameworks
Acceptance Criteria	Virtual user loops maintain stability thresholds under full target load constraints.
Related Requirements	FR17
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR18
Requirement Name	Modular Architecture
Requirement Description	The system architecture must be modular to allow independent feature additions. • Modules are loosely coupled with defined interfaces. • Adding or replacing a module must not require changes to unrelated modules.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Scalability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All modules
Acceptance Criteria	Component extraction avoids structural breaks within unrelated application features.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR19
Requirement Name	Scalable Storage
Requirement Description	Cloud image storage must scale automatically with no manual intervention required. • Storage uses an auto-scaling cloud object store (e.g. AWS S3). • No capacity cap warnings or manual provisioning are required during normal operation.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Scalability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR21
Acceptance Criteria	Bucket parameters accept scaling data payloads without capacity caps.
Related Requirements	FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR20
Requirement Name	Tech Documentation
Requirement Description	All API endpoints and key system logic must be documented. • API documentation is maintained in a standard format (e.g. OpenAPI / Swagger). • Documentation is updated with every significant code change.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Maintainability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All developers
Acceptance Criteria	Integration changes pass checklist tests validating corresponding documentation updates.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR21
Requirement Name	Cross-Browser Support
Requirement Description	The system must be compatible with Chrome, Safari, and Edge browsers. • Applies to the two most recent stable releases of each target browser. • UI layout and functionality must be consistent across all supported browsers.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Compatibility
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All interface views
Acceptance Criteria	UI layout tags load identically across all target vendor rendering engines.
Related Requirements	FR14
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR22
Requirement Name	Error Message Clarity
Requirement Description	Error messages must provide clear and actionable instructions for resolution. • Validation failure states output descriptive user-friendly messages. • Raw error codes or stack traces must never be exposed to end users.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR45
Acceptance Criteria	Validation failure states output descriptive strings rather than raw error codes.
Related Requirements	FR45
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR23
Requirement Name	Log Retention
Requirement Description	System logs must be retained for at least 90 days. • Logs older than 90 days may be archived or deleted by automated routines. • Retention policy applies to access logs, error logs, and audit logs.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Could
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR37
Acceptance Criteria	Routine system cleanup commands delete only log lines matching the expiry parameter.
Related Requirements	FR37
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR24
Requirement Name	Database Consistency
Requirement Description	The system must ensure ACID properties for all database transactions. • All write operations involving financial data use ACID-compliant transactions. • Aborted transactions roll back to their previous consistent state.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR17
Acceptance Criteria	Aborted payment actions roll back database values cleanly to their original state.
Related Requirements	FR17, FR25, FR44
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR25
Requirement Name	API Latency
Requirement Description	External API calls such as Google Maps must respond within 500 milliseconds. • Timeout thresholds are configured on all external API integrations. • Calls exceeding 500 ms are logged and trigger a graceful fallback.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Could
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR10, FR16
Acceptance Criteria	Handler integration points log or timeout tasks passing the 500 ms bound.
Related Requirements	FR16, FR39
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR26
Requirement Name	Password Complexity
Requirement Description	The system must enforce strong password requirements. • Minimum 8 characters. • Must include at least one uppercase letter, one lowercase letter, and one digit. • Trivial or common passwords are rejected at registration and reset.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR01, FR03
Acceptance Criteria	Registration rejects passwords that do not meet complexity requirements.
Related Requirements	FR01
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR27
Requirement Name	Localization Support
Requirement Description	Date and time formats should be localized based on the user's region settings. • Date and time formats adapt to the user's selected locale. • Supported locales: Arabic (Egypt) and English (UK/US).
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Could
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR38
Acceptance Criteria	Structural timestamps swap formats gracefully between target user locales.
Related Requirements	FR38
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR28
Requirement Name	Input Sanitization
Requirement Description	All user inputs must be sanitized to prevent SQL injection and XSS attacks. • Sanitization is applied server-side before any input reaches the database. • Script element injections are escaped before hitting database operations.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR04, FR21
Acceptance Criteria	Script element injections escape safely before reaching database operations.
Related Requirements	FR19
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR29
Requirement Name	Horizontal Scaling
Requirement Description	The system should support horizontal scaling across multiple servers. • The application layer must be stateless to support horizontal scaling. • A load balancer distributes traffic across available server nodes.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Scalability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Server infrastructure
Acceptance Criteria	Load balancer routing structures split compute stress across added server nodes.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR30
Requirement Name	Code Versioning
Requirement Description	All system code must be maintained in a version control system using Git. • All source code resides in a Git repository. • Commit messages must follow a defined convention.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Maintainability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Development lifecycle
Acceptance Criteria	Structural codebase commits pass alignment validations in the master repository.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR31
Requirement Name	Network Throttling
Requirement Description	The system must handle network throttling gracefully on mobile devices. • Progressive loading and skeleton screens are used while content is fetching. • Application must not crash on slow or interrupted connections.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All web interface views
Acceptance Criteria	Connectivity transitions prevent hard crashes via structural fallback loading modes.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR32
Requirement Name	HTTPS Enforcement
Requirement Description	All traffic must be served over HTTPS with a valid SSL certificate. • Plain HTTP requests are automatically redirected to HTTPS. • SSL certificates must be kept valid and auto-renewed.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All functions
Acceptance Criteria	Plain insecure server routes rewrite automatically into secure HTTPS structures.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR33
Requirement Name	Performance Monitoring
Requirement Description	The system should collect load time metrics for ongoing performance monitoring. • Instrumentation captures API response times and page load durations. • Metrics are accessible in a monitoring dashboard for Admins.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Backend layer
Acceptance Criteria	Instrumentation hooks capture operational timelines across all API tracks.
Related Requirements	FR31
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR34
Requirement Name	Zero-Downtime Deployment
Requirement Description	The system should support zero-downtime updates using rolling deployment. • New versions are deployed via a rolling update strategy. • User sessions must not be dropped during a deployment.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Production CI/CD pipelines
Acceptance Criteria	Version changes swap traffic pathways without severing ongoing user sessions.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR35
Requirement Name	Consistent Branding
Requirement Description	The UI must maintain consistent MSA University branding and color palette across all pages. • Official palette: Gold #8B6914 and Navy #1F3864. • Branding guidelines apply to all public-facing and dashboard screens.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All UI layouts
Acceptance Criteria	Gold and navy assets align with official MSA University campus guidelines on all pages.
Related Requirements	FR48
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR36
Requirement Name	Third-Party Privacy
Requirement Description	The system must not share user data with any third-party advertisers. • No advertising SDK or pixel tracking is embedded in the platform. • Privacy policy clearly states no data is sold or shared for advertising.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR01
Acceptance Criteria	Analytical track configurations exclude user data parameters from any advertiser network frames.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR37
Requirement Name	Search Precision
Requirement Description	Search results must have a precision rate of at least 95%. • Precision is measured as the proportion of returned results that match all applied filters. • Validated through automated test suites using predefined filter combinations.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR07, FR09
Acceptance Criteria	Result indices exclude noise data rows consistently across all testing suites.
Related Requirements	FR09, FR29
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR38
Requirement Name	Offline Support
Requirement Description	The mobile app should provide basic offline access to saved and shortlisted listings. • Shortlisted listings are cached locally for offline viewing. • No new search or communication is possible without connectivity.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Won't
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR13
Acceptance Criteria	Service worker caches load shortlisted arrays cleanly under total connectivity loss.
Related Requirements	FR13
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR39
Requirement Name	Resource Monitoring
Requirement Description	Admins must be automatically alerted if CPU usage exceeds 80%. • CPU usage is monitored continuously on all production server nodes. • Alerts are dispatched via email or SMS when the 80% threshold is crossed.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Could
Business Area	Reliability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All components
Acceptance Criteria	Threshold transitions activate notification relays to system engineers immediately.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR40
Requirement Name	Automated Testing
Requirement Description	The system should maintain at least 80% automated unit test coverage. • Coverage is measured by an automated tool integrated into the CI/CD pipeline. • Deployments are blocked if coverage drops below 80%.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Maintainability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Test suites
Acceptance Criteria	Coverage engine analytics reject deployments when coverage drops below 80%.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR41
Requirement Name	Legal Compliance
Requirement Description	The system must comply with local Egyptian student housing regulations. • Privacy controls match student data protection requirements under applicable Egyptian law. • Contract templates comply with local rental agreement regulations.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Legal
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All system policies
Acceptance Criteria	Privacy controls match student data protection requirements in the local legal framework.
Related Requirements	FR43
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR42
Requirement Name	Font Legibility
Requirement Description	Body text must be set to a minimum of 14px for readability on all screen sizes. • Applies to all body text elements across all screens. • Automated style checks flag any element with a font size below 14px.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Usability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All screens
Acceptance Criteria	Compiled layout style rules isolate and block any font-size declarations below 14px.
Related Requirements	FR04
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR43
Requirement Name	Low Latency OTP
Requirement Description	OTP messages must be delivered within 15 seconds of being requested. • Measured from API trigger to SMS delivery confirmation. • Applies under standard cellular network conditions.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR02, FR06
Acceptance Criteria	Cellular gateway loops clear text OTP payloads into network targets within 15 seconds.
Related Requirements	FR02, FR34
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR44
Requirement Name	Memory Management
Requirement Description	The application must not exceed 200 MB RAM usage on mobile devices. • Measured via browser performance profiling on a mid-range Android or iOS device. • Memory leak detection is part of the standard test suite.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	Mobile browser frameworks
Acceptance Criteria	Memory leak profiles remain flat within runtime monitoring profiles.
Related Requirements	FR24
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR45
Requirement Name	Secure File Upload
Requirement Description	All uploaded images must be scanned for malware before being stored. • Malware scanning runs server-side before any file is committed to storage. • Infected files are rejected and the uploader is notified with an error.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR21, FR27
Acceptance Criteria	Infected binary objects fail the parsing execution flow at the validation checkpoint.
Related Requirements	FR21
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR46
Requirement Name	API Rate Limiting
Requirement Description	The system must limit API requests per IP address to prevent denial-of-service attacks. • Rate limits are enforced on all public-facing API endpoints. • Requests exceeding the threshold receive a 429 Too Many Requests response.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All endpoints
Acceptance Criteria	Bursting calls past thresholds trigger 429 restriction responses.
Related Requirements	FR35
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR47
Requirement Name	Database Indexing
Requirement Description	Frequently searched query fields must be indexed for fast query execution. • Indexed fields include: price, gender, room type, listing location, and availability. • Index health is monitored and rebuilt when degraded.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Performance
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR07, FR10
Acceptance Criteria	Scanning queries trace balanced structural index keys instead of performing full table scans.
Related Requirements	FR08, FR42
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR48
Requirement Name	Clean Code Standards
Requirement Description	Code must follow an equivalent industry-standard style guide. • Python code follows PEP8. • JavaScript/TypeScript code follows the Airbnb or equivalent style guide. • Style compliance is enforced via linters in the CI/CD pipeline.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Maintainability
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	All microservices
Acceptance Criteria	Compilation tests flag any code syntax deviations from the configured style guide.
Related Requirements	None
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR49
Requirement Name	User Consent
Requirement Description	The system must obtain explicit user consent for all data collection. • A clear consent screen is presented during registration. • Data persistence routines remain dormant until the user confirms consent. • Consent records are stored with a timestamp for audit purposes.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Must
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR01
Acceptance Criteria	Persistence routines stay dormant until the interface consent confirmation loop fires.
Related Requirements	FR49
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

Requirement Identifier	NFR50
Requirement Name	Session Concurrency
Requirement Description	Users shall be limited to 3 active sessions simultaneously. • A fourth concurrent login terminates the oldest active session token. • The displaced session receives a “session expired” notification.
Source	MSA University CS347 Software Engineering Board
Owner	Faculty of Computer Science / Academic Supervisors
Author	Farah Kinaz, Retag Ahmed, Shahd Maher, Salah Bassem, Muhammad Kandil, Ahmed Walid
Type of Requirement	Non-Functional Requirement
Priority	Should
Business Area	Security
Stakeholders	Students, Parents, Housing Brokers, University Administrators
Associated Non-Functional Requirements	FR03
Acceptance Criteria	Initialization of a fourth concurrent login token terminates the oldest active session link.
Related Requirements	FR03
Related Documents	DorMsa Complete Requirements Catalogue v1.0, Project Charter
Comments	Evaluated against MSA University Housing compliance guidelines.
Version History	v1.0 – Baseline Specification Document
Pattern Name	Stephen Withall Requirement Evaluation Pattern (2007)

8 Preliminary Object-Oriented Domain Analysis

8.1 Inheritance Relationships

User is an abstract base class. Student, Broker, and Admin are concrete subclasses that inherit common attributes and operations such as userID, name, phone, login(), and logout(), while each adds its own role-specific behaviour.

```
User (abstract)
  |-- Student
  |-- Broker
  |-- Admin
```

Additional associations: Listing is owned by Broker. Booking links Student and Listing. Review is authored by Student and references a Listing. Notification can be sent to any User.

8.2 Class Descriptions

8.2.1 User (Abstract)

Purpose: Base entity representing any registered platform user.
Attributes: userID: String, name: String, phone: String (encrypted), role: Enum {Student, Broker, Admin}, isVerified: Boolean, sessionToken: String
Operations: register(), login(), logout(), editProfile(), resetPassword()

8.2.2 Student (Concrete)

Purpose: Represents an MSA student or parent looking for housing.
Attributes: studentID: String, savedSearches: List<SearchCriteria>, shortlist: List<Listing>, paymentHistory: List<Transaction>
Operations: searchListings(filters), saveListing(listing), payDeposit(listing, amount), submitRevie
rating, text)

8.2.3 Broker (Concrete)

Purpose: Represents a property owner or agent who manages listings.
Attributes: brokerID: String, idDocuments: List<File>, isApproved: Boolean, verificationBadge: Boolean, dashboardMetrics: DashboardData
Operations: createListing(), editListing(listing), toggleAvailability(listing), viewDashboard(), submitIdentity(docs)

8.2.4 Admin (Concrete)

Purpose: MSA staff member responsible for moderation and oversight.
Attributes: adminID: String, permissions: List<String>
Operations: verifyBroker(broker), moderateContent(listing), suspendUser(user), viewAnalytics()

8.2.5 Listing (Concrete)

Purpose: Represents a housing property posted by a verified Broker.
Attributes: listingID: String, brokerID: String, title: String, price: Float, roomType: Enum, gender: Enum, location: GeoPoint, photos: List<URL>, isAvailable: Boolean, createdAt: DateTime, updatedAt: DateTime
Operations: publish(), hide(), flagAsDuplicate(), getDistance(campusLocation)

8.2.6 Booking (Concrete)

Purpose: Represents a confirmed housing reservation between a Student and a Broker.
Attributes: bookingID: String, studentID: String, listingID: String, depositAmount: Float,

status: Enum {Pending, Confirmed, Cancelled}, confirmedAt: DateTime
Operations: confirmBooking(), cancelBooking(), generateReceipt()

8.2.7 Review (Concrete)

Purpose: Stores student feedback for a listing or broker.
Attributes: reviewID: String, authorID: String, targetID: String, rating: Integer (1--5), text: String, createdAt: DateTime
Operations: submit(), publish(), delete()

8.2.8 Notification (Concrete)

Purpose: System-generated message dispatched to users on trigger events.
Attributes: notifID: String, recipientID: String, type: Enum {Booking, Expiry, NewListing}, channel: Enum {SMS, Push, Email}, deliveredAt: DateTime
Operations: send(), retry(), markDelivered()

9 Preliminary Schedule

- **Phase 1 — Requirements Analysis (Weeks 1–2):** Stakeholder requirements gathering and SRS documentation finalisation. *Milestone: SRS Document Approved & Signed Off.*
- **Phase 2 — System Design (Weeks 3–4):** Diagrams, data modelling, class diagrams, and system architecture. *Milestone: Architecture Approved.*
- **Phase 3 — Core Development (Weeks 5–9):** User registration APIs, multi-parameter search and filters, broker property upload, and connecting the search interface to backend services. *Milestone: Core Housing Search Engine Functional.*
- **Phase 4 — Advanced Features & Management (Weeks 10–12):** Admin moderation panels, platform control setup, and analytical performance dashboards. *Milestone: Feature-Complete Integration Freeze.*

10 Preliminary Budget

Item	Estimated Cost (USD)
Cloud Hosting — AWS EC2 + S3 (12 months)	$240 / year
SMS Gateway — Twilio OTP delivery	$50 / 1,000 messages
Payment Gateway — Stripe or PayMob	1.5% per transaction
Map Services — Google Maps API	$200 / year
SSL Certificate — Let's Encrypt	Free
Development Tools — GitHub, Figma (student plans)	$0
Total estimated annual operating cost	**˜$500 / year**

11 Appendices

11.1 Appendix A — Use Case Diagram

The DorMsa Use Case Diagram identifies four actors: General User, Student/Parent, Broker, Admin, and System (Automated Tasks). The main use cases are: Register Account (includes Verify Phone Number), Search and Filter Listings, Manage Listings, Verify Brokers, Moderate Content, View Analytics, and Send Notifications.

11.2 Appendix B — Requirements Traceability Matrix

Business Goal	Mapped Requirements
Ensure User Trust and Security	FR: FR01, FR02, FR03, FR27, FR28, FR30, FR37, FR48, FR49 NFR: NFR01, NFR03, NFR04, NFR05, NFR26, NFR28, NFR32, NFR45, NFR46, NFR49
Enable Effective Property Search	FR: FR07, FR08, FR09, FR10, FR11, FR13, FR16, FR20, FR40, FR41 NFR: NFR10, NFR14, NFR25, NFR37, NFR47
Facilitate Broker-Student Communication	FR: FR14, FR15, FR24, FR25 NFR: NFR07, NFR21, NFR31
Manage Listings and Inventory	FR: FR21, FR22, FR23, FR26, FR33, FR36, FR47 NFR: NFR09, NFR19, NFR24
Secure Financial Transactions	FR: FR17, FR34, FR43, FR44 NFR: NFR05, NFR01, NFR36, NFR49
Provide Admin Oversight	FR: FR28, FR29, FR30, FR31, FR32, FR37 NFR: NFR03, NFR06, NFR23, NFR39
Deliver Notifications and Alerts	FR: FR34, FR35, FR36, FR46 NFR: NFR13, NFR43
Ensure Performance and Scalability	FR: FR10, FR17, FR47 NFR: NFR08, NFR09, NFR10, NFR13, NFR15, NFR16, NFR17, NFR19, NFR29, NFR34, NFR43, NFR44, NFR47
Maintain System Quality and Compliance	FR: FR37, FR43, FR50 NFR: NFR12, NFR18, NFR20, NFR24, NFR30, NFR40, NFR41, NFR48
Provide User Support and Accessibility	FR: FR38, FR39, FR42, FR45, FR50 NFR: NFR07, NFR11, NFR12, NFR22, NFR27, NFR35, NFR38, NFR42

11.3 Appendix C — Requirement Priority Summary

- **Must:** 19 Functional, 33 Non-Functional — mandatory; the system cannot launch without these.
- **Should:** 15 Functional, 14 Non-Functional — high priority, strongly desired for initial release.
- **Could:** 16 Functional, 3 Non-Functional — optional, planned for future phases.
- **Won't:** 2 Functional (FR31, FR32), 1 Non-Functional (NFR38) — explicitly out of scope for this release.

11.4 Collected Material

- DorMsa Complete Requirements Catalogue (v1.0) — source of all 89 requirements.
- Phase 3 SRS Draft — use case diagram and scenario descriptions.

- MSA CS SRS Template (Overleaf) — structural guideline.

12 Diagrams

The following diagrams are referenced from the project artefacts. Insert image files as needed using \includegraphics once compiled locally.

12.1 Use Case Diagram

[Insert Use Case Diagram here]

12.2 Sequence Diagram

[Insert Sequence Diagram here]

12.3 Class Diagram

[Insert Class Diagram here]

12.4 Rich Picture

[Insert Rich Picture here]

12.5 Gantt Chart

[Insert Gantt Chart here]

13 References

1. MSA University, “CS347 Software Engineering — SRS Project Guidelines”, Faculty of Computer Science, 2024.
2. IEEE Std 830-1998, “IEEE Recommended Practice for Software Requirements Specifications”.
3. I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2016.
4. PCI Security Standards Council, “PCI DSS v4.0”, 2022.
5. W3C, “Web Content Accessibility Guidelines (WCAG) 2.1”, 2018.
6. S. Withall, *Software Requirement Patterns*, Microsoft Press, 2007.